

Individual Honours Project

WEEKLY REPORT

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UIT - BCU

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I. SUMMARY

- This is the report of my project “**BCG: A Game-Based Approach to Motor Imagery Adaptation and BCI Calibration.**”
- The target was to make EEG-based motor imagery calibration more effective and easier to apply outside the laboratory.
- Current progress:
 - Found a suitable public dataset for model training
 - Investigating Python libraries for signal processing
 - Wrote the abstract for the BME 11 conference

II. What I have done

Model Dataset: MIMED

- Published: 1 August 2024
- Link: <https://data.mendeley.com/datasets/zs25xxjkm9/3>
- Device: Emotiv EPOC X (14 channels)
- Subjects: 30 students (Bali, Indonesia)
- Sessions: 2 days, 4 scenarios/day
- Duration: 8 mins/scenario
- Data type: Raw EEG + feature data

II. What I have done

Investigating Python libraries for signal processing

The screenshot displays the Visual Studio Code (VS Code) interface. On the left, the Explorer sidebar shows a project named 'MODEL' with a subfolder 'MIMED'. Inside 'MIMED', there are several files including 'Convert EDF-to-Mat', 'Dataset_1D', 'Matlab Dataset', 'Motor Execution', 'Motor Imagery', 'Supplementary_file', 'Supplementary_file 2', '1D_Data - Execution.py', '1D_Data - Imagery.py', 'Convert EDF-to-Mat.zip', 'Dataset_1D.zip', 'DecisionTree_6Class.py', 'Matlab Dataset.zip', 'Media Stimulus English Ver.mp4', 'Motor Execution.zip', 'Motor Imagery.zip', 'Supplementary_file.zip', 'SVM_6Class.py', 'venv', 'load_mat.py', and 'MIMED.zip'. The file 'load_mat.py' is selected and open in the main editor. The code in 'load_mat.py' is as follows:

```
11
12 data = sio.loadmat("MIMED/Motor Imagery/Left Hand Up-Down Imagine/P01.mat")
13
14 print("Fs:", data['Fs'][0][0])
15 print("Channels:", [str(c[0]) for c in data['channel'][0]])
16 print("Subject:", data['subject'][0])
17
18
19 for i in range(4):
20     scenario = data['joined_data'][0][i]
21     print(f"\nScenario {i+1}: type={type(scenario)}, shape={scenario.shape}")
22
```

Below the code editor, the TERMINAL panel shows the execution of the script. The terminal output is as follows:

```
(venv) dkhoamac@Khoas-MacBook-Pro model % pip install scipy
(venv) dkhoamac@Khoas-MacBook-Pro model % python load_mat.py
Fs: type=<class 'numpy.ndarray'>, shape=(1, 1)
channel: type=<class 'numpy.ndarray'>, shape=(1, 14)
joined_data: type=<class 'numpy.ndarray'>, shape=(1, 4)
subject: type=<class 'numpy.ndarray'>, shape=(1,)
(venv) dkhoamac@Khoas-MacBook-Pro model % pip install numpy
Requirement already satisfied: numpy in ./venv/lib/python3.13/site-packages (2.4.3)
[notice] A new release of pip is available: 24.3.1 -> 26.0.1
[notice] To update, run: pip install --upgrade pip
(venv) dkhoamac@Khoas-MacBook-Pro model % python load_mat.py
Fs: type=<class 'numpy.ndarray'>, shape=(1, 1)
channel: type=<class 'numpy.ndarray'>, shape=(1, 14)
joined_data: type=<class 'numpy.ndarray'>, shape=(1, 4)
subject: type=<class 'numpy.ndarray'>, shape=(1,)
Fs: 128
Channels: ['AF3', 'F7', 'F3', 'FC5', 'T7', 'P7', 'O1', 'O2', 'P8', 'T8', 'FC6', 'F4', 'F8', 'AF4']
Subject: P01

Scenario 1: type=<class 'numpy.ndarray'>, shape=(2432, 14)
Scenario 2: type=<class 'numpy.ndarray'>, shape=(2432, 14)
Scenario 3: type=<class 'numpy.ndarray'>, shape=(2432, 14)
Scenario 4: type=<class 'numpy.ndarray'>, shape=(2432, 14)
(venv) dkhoamac@Khoas-MacBook-Pro model %
```

The status bar at the bottom of the VS Code window shows the current file is 'main*', there are 0 errors and 2 warnings, and the file is being live shared to AWS. The bottom right corner shows the current line and column as 'Ln 18, Col 1', the file encoding as 'UTF-8', the file type as 'Python', the Python version as '3.13.1', and the virtual environment as 'venv (3.13.2)'.

IV. Next Steps

- Finish the model by the deadline (29 Mar)
- Finalise the BME 11 abstract then submit



THANK YOU
FOR READING!

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